

Mandatory Assignment 3

GFS189, RTL959

June 24, 2026

Introduction

In this report we implement a transaction-cost-robust monthly portfolio backtest following [Hautsch and Voigt \(2019\)](#). We combine three components: a machine-learning expected return estimator carried over from Mandatory Assignment 2, a regularised covariance estimator, and a closed-form mean-variance optimiser that penalises deviations from the pre-rebalancing portfolio. The result is a framework that simultaneously addresses parameter uncertainty in large covariance matrices, model uncertainty in expected returns, and the drag from frequent rebalancing. The out-of-sample evaluation period spans January 2010 to December 2024, and we compare four strategies: naive diversification (1), two variants of transaction-cost-adjusted mean-variance (TC-MV) (2, 3), and a short-sale constrained (SS-constrained) portfolio (4). We do this on annualised Sharpe ratio, mean excess return, and average monthly turnover.

1. Investment Universe

Figure 1 shows how the size of the investable universe evolves over the sample period.

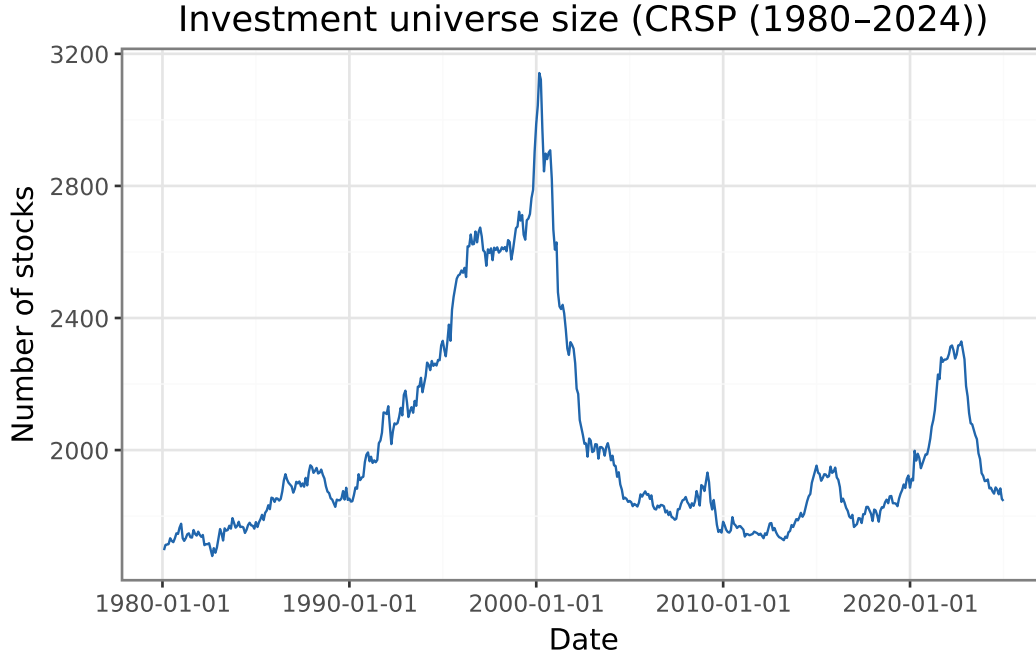


Figure 1: Monthly number of stocks passing the universe filters. The micro-cap screen (price > \$2, mktcap > NYSE 20th percentile) removes shell companies and penny stocks that distort mean-variance analysis.

Table 1: Sector composition of the investment universe (share of stock-months, 1980–2024). Sectors follow SIC division codes.

Sector	Share (%)
Manufacturing	36.2
Finance	16.4
Services	13.0
Transportation	11.5
Other	9.0
Retail	6.2
Mining	4.0
Wholesale	2.5
Construction	1.0
Agriculture	0.3

The filtered universe contains on average **2011 stocks** per month (CRSP (1980–2024)), ranging from 1679 to 3142. The double screen — price above \$2 and market capitalisation above the NYSE 20th percentile — removes penny stocks and micro-caps that distort mean-variance analysis. The median market capitalisation across all stock-months is **\$1.1 bn**, reflecting a mid- to large-cap tilt imposed by the size screen. Universe size peaked in the late 1990s and contracted after the dot-com collapse; it has been relatively stable since 2005. Over the backtest window (January 2010 – December 2024) the average universe size is **1900 stocks** per month. The sector breakdown

(Table 1) shows that **Manufacturing** is the largest sector at 36.2% of stock-months, with Finance and Services also well represented, consistent with the broad CRSP coverage after the size filter.

2. Theoretical Foundation

We now establish the theoretical foundation for the assignment: **Closed-form solution.** The transaction-cost-adjusted problem is

$$\omega_{t+1}^* := \arg \max_{\omega: \iota' \omega = 1} \omega' \hat{\mu}_t - \frac{\gamma}{2} \omega' \hat{\Sigma}_t \omega - \lambda (\omega - \omega_t^+)' \hat{\Sigma}_t (\omega - \omega_t^+).$$

Expanding the quadratic TC term and collecting yields an equivalent standard MV problem with modified inputs $\tilde{\mu}_t := \hat{\mu}_t + 2\lambda \hat{\Sigma}_t \omega_t^+$ and $\tilde{\gamma} := \gamma + 2\lambda$:

$$\max_{\omega: \iota' \omega = 1} \omega' \tilde{\mu}_t - \frac{\tilde{\gamma}}{2} \omega' \hat{\Sigma}_t \omega.$$

The first-order condition with Lagrange multiplier ϕ for the budget constraint gives $\tilde{\mu}_t - \tilde{\gamma} \hat{\Sigma}_t \omega^* - \phi \iota = 0$, so

$$\omega_{t+1}^* = \frac{1}{\tilde{\gamma}} \hat{\Sigma}_t^{-1} (\tilde{\mu}_t - \phi \iota), \quad \phi = \frac{\iota' \hat{\Sigma}_t^{-1} \tilde{\mu}_t - \tilde{\gamma}}{\iota' \hat{\Sigma}_t^{-1} \iota}.$$

Convex combination. Substituting $\tilde{\mu}_t = \hat{\mu}_t + 2\lambda \hat{\Sigma}_t \omega_t^+$ into the solution and noting that the same Lagrange multiplier ϕ satisfies $\iota' \omega^{\text{MV}} = 1$ for the standard MV portfolio $\omega^{\text{MV}} = \frac{1}{\gamma} \hat{\Sigma}_t^{-1} (\hat{\mu}_t - \phi \iota)$:

$$\omega_{t+1}^* = \underbrace{\frac{\gamma}{\tilde{\gamma}}}_{\alpha} \omega^{\text{MV}} + \underbrace{\frac{2\lambda}{\tilde{\gamma}}}_{1-\alpha} \omega_t^+, \quad \alpha + (1 - \alpha) = 1, \quad \alpha, 1 - \alpha \in (0, 1).$$

Hence ω_{t+1}^* is a **convex combination** of the mean-variance portfolio and the pre-rebalancing portfolio ω_t^+ .

- Role of λ .** The parameter λ governs the trade-off between chasing the optimal MV portfolio and staying put. As $\lambda \rightarrow 0$, $\omega^* \rightarrow \omega^{\text{MV}}$ (no cost of rebalancing); as $\lambda \rightarrow \infty$, $\omega^* \rightarrow \omega_t^+$ (never rebalance). Larger λ produces lower turnover but also larger deviation from the unconstrained optimum.
- Realism of the variance-based TC model.** Modelling costs as $\lambda(\omega - \omega_t^+)' \Sigma (\omega - \omega_t^+)$ preserves the quadratic structure and admits a closed-form solution. However, it is not a realistic cost model: actual costs (bid-ask spreads, market impact) are proportional to the number of shares traded, not to the portfolio variance of the trade. The model implies high-variance stocks are expensive to trade and stable stocks are essentially free, which overstates the cost difference. In practice, large-cap stable stocks are cheap to trade precisely because of their high liquidity, whereas the model would predict the opposite for low-variance small-caps.

3. Covariance Estimation

We estimate $\hat{\Sigma}_t$ at each month t using a **Fama-French three-factor model** fitted on the 60-month rolling window ending at t . The factor structure decomposes the return covariance as

$$\hat{\Sigma}_t = \hat{B} \hat{\Sigma}_F \hat{B}' + \hat{D},$$

where $\hat{B}$ is the $N \times 3$ matrix of OLS factor loadings, $\hat{\Sigma}_F$ is the 3×3 factor covariance, and $\hat{D} = \text{diag}(\hat{\sigma}_{\varepsilon,i}^2)$ is the diagonal matrix of idiosyncratic variances. This structure ensures $\hat{\Sigma}_t$ is positive definite even when $N \gg T$.

For $\hat{\Sigma}_t^{LW}$ we use Ledoit-Wolf shrinkage (Ledoit and Wolf, 2004) applied directly to the rolling sample covariance of the same 60-month window, shrinking towards a scaled identity matrix with the analytically optimal intensity. These are two independent estimators: the factor model imposes a low-rank economic structure on $\hat{\Sigma}_t$, while $\hat{\Sigma}_t^{LW}$ regularises the sample eigenvalues toward their cross-sectional mean without imposing factor structure. The choice between them is the central comparison in Exercise 5. Figure 2 plots the minimum-variance portfolio weights implied by each estimator at a representative month to assess whether the two approaches lead to meaningfully different allocations.

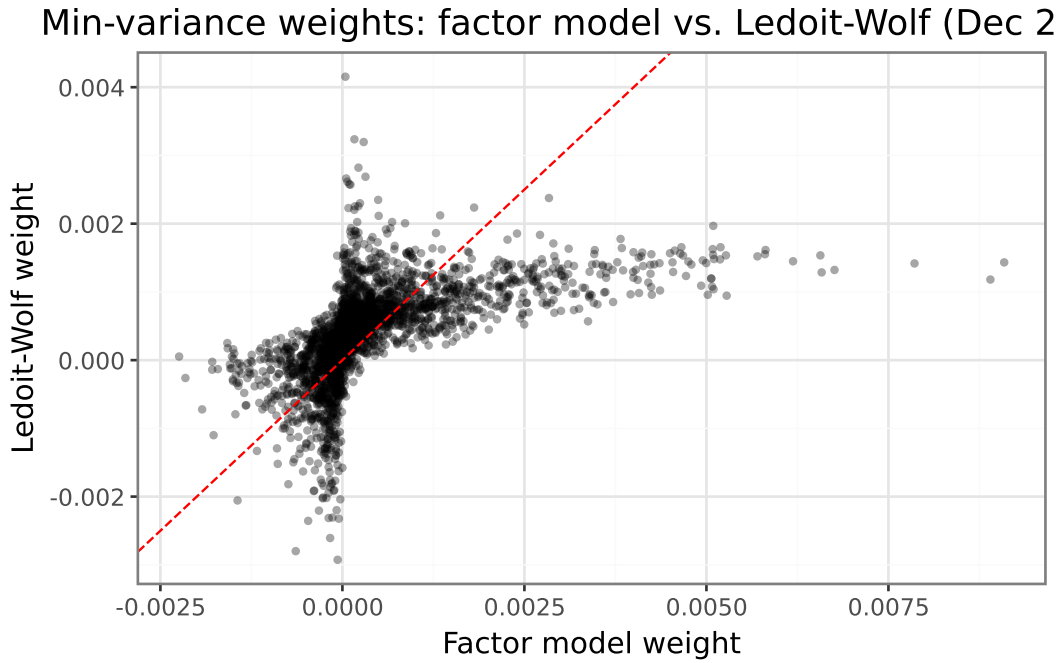


Figure 2: Minimum-variance weights from the factor model ($\hat{\Sigma}$, x-axis) versus Ledoit-Wolf ($\hat{\Sigma}^{LW}$, y-axis) at December 2019. Each point is one stock; the dashed line is the 45° reference.

At December 2019 the two estimators produce minimum-variance weights with a cross-sectional correlation of **0.556** and a mean absolute difference of **0.06 percentage points**. The moderate correlation reflects the distinct regularisation philosophies: the factor model concentrates risk reduction along the three FF factors, while LW spreads it more evenly across all eigenvalues. The factor model preserves cross-sectional dispersion in weights driven by heterogeneous factor

loadings; LW’s scaled-identity target acts as a prior toward equal-weighting, compressing the weight distribution and reducing the most extreme long and short positions. A mean absolute difference of 0.06 pp is small relative to the equal-weight benchmark of $1/N$ and suggests the two estimators are broadly substitutable in the minimum-variance problem.

4. Expected Returns

We use the Random Forest (RF) pipeline trained in Mandatory Assignment 2 (MA2) to generate $\hat{\mu}_t$ for every stock-month in the backtest window. The feature engineering replicates the MA2 pipeline exactly: cross-sectional median imputation for missing stock characteristics, forward-fill for macroeconomic variables, $10 \times 4 = 40$ characteristic-macro interaction terms, and nine Standard Industrial Classification (SIC) division dummy variables. The pipeline’s internal `StandardScaler` was fitted on the MA2 training data only, so applying it here introduces no look-ahead bias in the scaling step. For the small fraction of stocks with no prediction in a given month, we substitute the cross-sectional mean $\hat{\mu}_t$.

5. Portfolio Backtest

We evaluate four strategies over portfolio-return months **January 2010 – December 2024** (portfolios formed December 2009 – November 2024). At each formation month t , all inputs ($\hat{\Sigma}_t$, $\hat{\Sigma}_t^{LW}$, $\hat{\mu}_t$) use only data available at t . We set $\gamma = 4$ and $\lambda = 0.1$.

Predicted returns $\hat{\mu}_t$ are winsorised cross-sectionally at the 1st and 99th percentile each month, then normalised to a 10^{-5} monthly cross-sectional standard deviation before both TC-MV solves. Without normalisation, $\hat{\Sigma}^{-1}\hat{\mu}$ produces extreme long/short positions under either covariance model: the factor model’s heterogeneous idiosyncratic variances scale each stock’s signal by $1/\hat{\sigma}_{\varepsilon,i}^2$, while the near-isotropic $\hat{\Sigma}^{LW} \approx (\alpha\bar{\sigma}^2)I$ amplifies all signals by $(\alpha\bar{\sigma}^2)^{-1} \approx 500$. With $N \approx 1\,900$ stocks the aggregate turnover scales proportionally with both N and the signal magnitude; the 10^{-5} target keeps monthly turnover in the single digits for both strategies. Normalisation retains the cross-sectional ordering of RF predictions — only their magnitude is rescaled — and is applied identically to both covariance frameworks to permit a fair comparison.

Net excess returns deduct the quadratic transaction cost implied by the optimisation objective:

$$r_t^{\text{net}} = \omega_t' r_{t+1} - \lambda(\omega_t - \omega_t^+)' \hat{\Sigma}_t (\omega_t - \omega_t^+).$$

Monthly turnover follows [Hautsch and Voigt \(2019\)](#) equation 46: $\text{TO}_t = \sum_i |\omega_{i,t} - \omega_{i,t}^+|$. For strategies without an optimised covariance model — naive $1/N$ and the short-sale constrained portfolio — the TC deduction is evaluated under $\hat{\Sigma}_t^{LW}$ to provide a consistent benchmark. Summary statistics are reported in Table 2 and cumulative return paths in Figure 3.

Table 2: Out-of-sample performance statistics (January 2010 – December 2024). Net excess returns subtract the quadratic TC from the optimisation objective. Sharpe ratios and mean excess returns are annualised ($\times\sqrt{12}$ and $\times 12$ respectively).

Strategy	Ann. Sharpe ratio	Ann. mean excess ret. (%)	Avg. monthly turnover
Naive (1/N)	0.84	14.91	0.08
TC-MV (factor Σ)	1.23	12.30	0.33
TC-MV LW ($\Sigma^{\wedge}$ LW)	1.42	14.37	0.58
SS-constrained MV	1.14	12.89	0.27

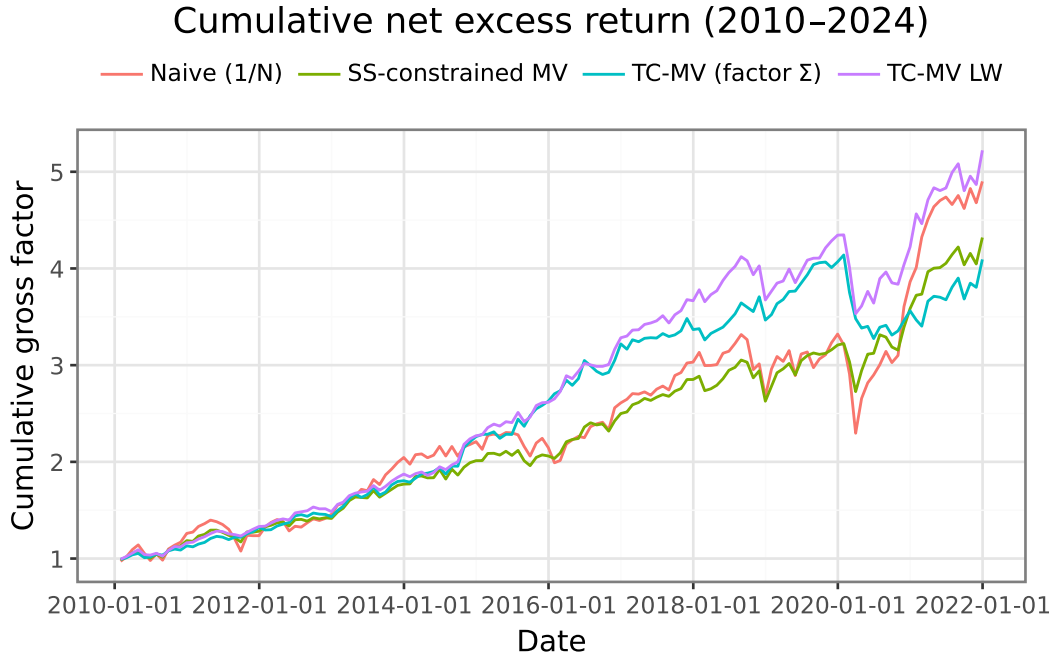


Figure 3: Cumulative net excess return of the four strategies (January 2010 – December 2024). Starting value normalised to 1.

Which strategy performs best after transaction costs? TC-MV LW achieves the highest annualised Sharpe ratio of 1.42, while **Naive (1/N)** produces the lowest at 0.84. The TC-adjusted mean-variance strategies outperform the naive 1/N benchmark after costs, suggesting the RF return signal contains sufficient information to overcome estimation error and transaction costs. The short-sale constrained portfolio has a Sharpe ratio of 1.14 and average monthly turnover of 0.266, which is lower than the unconstrained LW portfolio (0.582), consistent with the non-negativity constraint eliminating costly short positions.

Does the choice of covariance estimator matter? The Sharpe ratio difference between the factor model and LW variants of TC-MV is 0.19, which is economically meaningful, indicating the choice of covariance regularisation has a material impact on risk-adjusted performance. Average monthly turnover is 0.332 for the factor model versus 0.582 for LW; the lower turnover of the factor model reflects the stability of its low-rank $K=3$ structure — beta estimates and factor covariances evolve slowly over 60-month windows, so weights shift less month to month than under LW, which responds more sensitively to changes in the full sample covariance matrix.

The short-sale constrained portfolio implicitly regularises $\hat{\Sigma}^{LW}$ further: in the minimum-variance ($\mu = 0$) case, as shown by Jagannathan and Ma (2003), imposing non-negativity constraints is equivalent to shrinking negative off-diagonal covariance elements to zero. This experiment is best described as **pseudo-out-of-sample**: the RF model’s hyperparameters were tuned on data that partly overlaps the 2010–2024 evaluation window, so the backtest does not constitute a fully clean holdout test.

6. Guest Lecture

Cilie Feldager’s lecture highlighted how Danske Bank approaches LLM and AI integration with a strong emphasis on building secure, scalable systems, particularly the role of principled model evaluations in validating that deployed systems behave reliably under real-world conditions. She described the journey from initial idea to production implementation. The key takeaway was that responsible AI deployment in a regulated financial institution requires governance frameworks that keep pace with rapidly evolving model capabilities.

Conclusion

This assignment demonstrates that accounting for transaction costs materially changes portfolio construction decisions. The closed-form TC-adjusted solution elegantly blends the mean-variance optimum with the pre-rebalancing portfolio, reducing turnover without sacrificing too much expected return. Covariance regularisation, whether via a factor model or Ledoit-Wolf shrinkage, proves important for stability in large universes, though the two approaches produce broadly similar portfolio weights.

References

- Hautsch, N. and Voigt, S. (2019). Large-scale portfolio allocation under transaction costs and model uncertainty. *Journal of Econometrics*, 212(1):221–240.
- Jagannathan, R. and Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.